

SongNet: Real-Time Music Genre Classification from Raw Audio

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Code: <https://github.com/shruti956/songnet-music-genre>

1. Problem Statement

We classify music into 8 genres directly from raw audio, and do it in real time, so that a prediction is available while the song is still playing. The project is based on the SongNet paper (Stanford CS229, 2018).

2. Dataset

- **Source:** Free Music Archive, *fma_small* subset: 8,000 tracks, 30-second clips.
- **Classes:** 8 balanced genres, 1,000 clips each: Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, Rock.
- **Split:** stratified 70/20/10 (train/val/test) with seed 42: 5,600 / 1,600 / 800 clips. A few corrupted mp3s were skipped, leaving 5,599 / 1,599 / 799.
- **Labels and features:** labels come from tracks.csv; the baselines use the 518 precomputed features from features.csv.

3. Approach

- **Baselines:** k-NN, Logistic Regression, MLP and Linear SVM on the 518 precomputed FMA features (standardised).
- **Main model (C-RNN):** takes only a mel-spectrogram (128 mel bins, 22.05 kHz, 1,290 frames per clip, scaled to roughly [-1, 1]).
- **Architecture:** two 1-D convolution blocks along the time axis (kernel 5, 128 channels, batch norm, ReLU, max-pool 2, dropout 0.3), then a GRU (128 units), then a linear layer that outputs genre probabilities at every timestep.
- **Real-time prediction:** the song-level prediction is the average of the per-timestep predictions, which is what makes real-time classification possible.
- **Training:** Adam, lr 1e-3, batch size 64, 30 epochs, random 256-frame crops of each clip; the best checkpoint is chosen by validation accuracy.

4. Implementation Overview

The pipeline is: download, split, extract and cache spectrograms, train baselines, train the C-RNN, evaluate. It was run on Google Colab (CPU for extraction and baselines, GPU for training) using the notebook *notebooks/colab_runner.ipynb*. The code lives in *src/* (data, features, baselines, model, train, evaluate), alongside a Streamlit demo in *app/demo.py* that lets you upload a song and watch the prediction update over time. The repository is organised as README.md, requirements.txt, splits.csv, models/best.pt, src/, app/, notebooks/ and report/, and is available at the GitHub link above.

5. Results

Model	Test Accuracy
Random guessing	12.5%
k-NN	44.5%
Logistic Regression	54.5%
MLP	58.0%
Linear SVM	53.5%
C-RNN	60.2%

Table 1: Test accuracy (799 test clips).

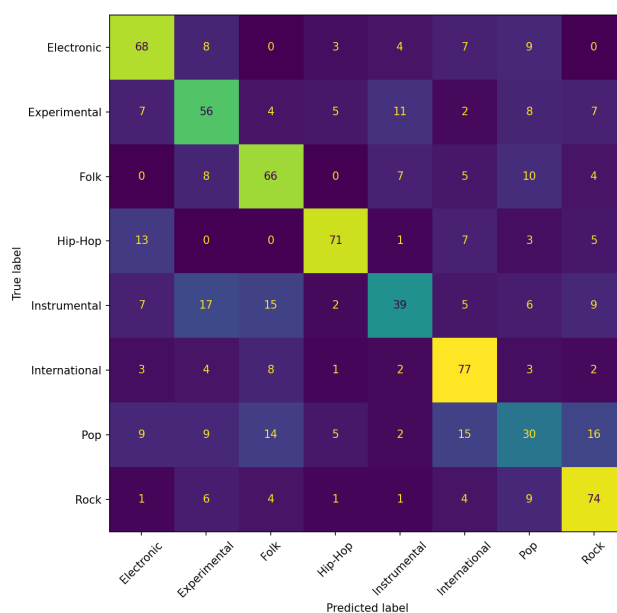


Figure 1: C-RNN confusion matrix on the test set.

6. Conclusions

- The C-RNN achieves the best test accuracy (60.2%), but only about 2 points above the best baseline (MLP, 58.0%), so it slightly outperforms the baselines.
- It does this from the raw mel-spectrogram alone, while the baselines use 518 handcrafted features, and it provides real-time predictions.
- **Best classes:** International, Rock, Hip-Hop, Electronic, Folk. **Hardest:** Pop (confused with Rock, International, Folk) and Instrumental (confused with Experimental, Folk).
- **Limitations:** a small dataset (fma_small), a single run with no repeats, and a small gap between the C-RNN and the MLP on an 800-clip test set.
- **Future work:** the larger FMA subsets, data augmentation, and deeper or bidirectional recurrent layers.